

ST PAUL AGCM, AB, Canada

WMO: 712750

Lat: 54.0111N

Lon: 111.2719W

Elev: 2129

StdP: 13.60

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.5	-22.3	-34.8	0.8	-28.3	-28.3	1.2	-22.1	21.2	18.5	19.1	17.3	3.2	230	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	81.5	63.4	78.2	62.1	75.2	60.9	66.5	77.5	64.3	74.7	62.4	72.1	8.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	90.9	71.8	60.2	84.4	69.3	58.5	79.1	66.8	32.3	77.4	30.6	74.6	29.2	72.4	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	18.2	16.1	DB	-38.0	86.5	5.3	2.3	-41.8	88.1	-44.9	89.5	-47.9	90.8	-51.7	92.4
			WB	-37.5	71.0	5.2	1.8	-41.2	72.3	-44.3	73.3	-47.2	74.3	-51.1	75.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.9	7.5	7.2	20.3	36.3	50.2	57.7	62.0	59.1	50.3	37.9	20.6	8.0
	DBStd	22.99	15.69	13.90	13.34	9.80	8.02	5.63	5.10	5.90	8.03	8.60	12.60	14.45
	HDD50	6627	1319	1198	920	421	98	8	0	5	92	384	881	1301
	HDD65	11035	1784	1618	1385	862	461	226	122	197	443	840	1331	1766
	CDD50	1122	0	0	0	9	103	240	373	287	101	9	0	0
	CDD65	55	0	0	0	0	2	8	29	14	2	0	0	0
	CDH74	904	0	0	0	2	69	129	390	222	89	3	0	0
	CDH80	153	0	0	0	0	8	14	83	31	17	0	0	0
Wind	WSAvg	7.4	7.3	6.8	7.7	8.6	8.4	7.7	6.8	6.7	7.3	7.7	7.3	6.8
Precipitation	PrecAvg	17.2	0.7	0.6	0.7	1.2	1.7	3.0	3.3	2.4	1.4	0.9	0.8	0.7
	PrecMax	20.7	2.0	1.3	1.4	3.4	3.2	6.4	7.2	5.4	3.6	2.2	2.2	1.4
	PrecMin	10.9	0.0	0.0	0.2	0.1	0.2	0.9	1.3	0.9	0.1	0.2	0.1	0.0
	PrecStd	2.6	0.4	0.3	0.4	0.7	0.8	1.3	1.1	1.0	0.9	0.4	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.0	40.7	50.5	70.2	80.9	82.1	86.4	84.1	83.1	71.7	52.1	40.7
		MCWB	36.1	34.8	41.2	50.9	57.2	61.7	67.0	65.4	60.7	53.8	42.1	35.7
	2%	DB	37.1	35.7	44.0	62.7	75.2	77.8	82.1	79.2	75.8	61.5	44.1	36.0
		MCWB	33.3	31.8	37.5	47.1	55.0	60.5	65.7	63.5	58.8	49.0	38.4	32.2
	5%	DB	33.5	31.7	40.4	56.9	70.6	73.9	78.4	76.1	70.5	56.0	39.2	31.9
		MCWB	30.9	29.1	35.0	44.0	52.7	58.4	64.0	62.7	56.7	46.2	34.6	29.1
	10%	DB	29.6	27.7	37.1	52.0	66.7	70.5	74.8	72.9	65.2	51.5	35.7	27.7
		MCWB	27.3	25.4	32.9	41.5	50.6	56.8	62.5	61.0	53.9	43.3	32.5	25.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	35.7	42.0	52.0	59.2	65.2	70.2	68.9	62.5	54.6	43.5	36.3
		MCDB	40.4	39.9	49.2	67.9	74.6	76.2	80.4	78.5	78.3	69.6	51.1	40.7
	2%	WB	33.8	32.1	37.6	48.1	56.1	62.5	67.6	65.7	59.9	50.4	39.0	32.5
		MCDB	36.5	35.1	43.2	60.8	70.8	72.8	79.1	76.1	73.1	60.0	43.9	35.9
	5%	WB	31.0	28.9	35.3	44.7	54.1	60.5	65.4	63.6	57.4	46.8	35.3	29.3
		MCDB	33.6	31.8	39.6	55.9	67.9	70.5	76.2	73.6	69.0	54.2	38.5	31.8
	10%	WB	27.4	25.2	33.4	42.1	52.0	58.6	63.3	61.7	54.9	44.3	32.6	25.6
		MCDB	29.6	27.4	37.0	51.2	63.8	67.9	72.9	70.8	63.8	51.0	35.3	27.7
Mean Daily Temperature Range	5% DB	MDBR	17.9	19.7	18.6	19.3	23.0	20.3	21.4	21.9	22.0	18.9	15.1	16.3
		MCDBR	18.8	22.7	21.2	27.7	29.0	26.2	27.2	27.6	31.1	27.6	18.9	20.0
		MCWBR	15.8	19.5	15.5	15.1	13.2	12.8	13.9	14.3	15.9	16.5	13.5	17.0
	5% WB	MCDBR	18.6	23.0	19.5	26.3	26.1	22.3	24.4	24.2	28.4	25.8	18.0	20.2
		MCWBR	15.9	19.5	14.3	15.1	12.4	11.9	13.2	13.4	15.0	16.1	13.3	17.2
Clear-Sky Solar Irradiance	taub	0.250	0.270	0.289	0.327	0.345	0.361	0.360	0.369	0.308	0.275	0.249	0.223	
	taud	2.295	2.279	2.348	2.375	2.377	2.351	2.392	2.330	2.502	2.565	2.448	2.331	
	Ebn at Noon	232	265	285	287	286	281	279	269	277	266	237	224	
	Edh at Noon	18	25	30	34	36	37	35	35	26	19	15	14	
All-Sky Solar Radiation	RadAvg	290	621	1104	1481	1823	1864	1908	1555	1093	618	291	208	
	RadStd	31	48	82	126	141	132	103	119	118	94	35	20	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page